

Program 1 :

Docker file :

```
Nov 3 20:48
abhi@abhi-ThinkPad-T490: ~/Desktop/DevopsLab/program1

abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program1$ ls
Dockerfile  node_modules  package.json  package-lock.json  server.js
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program1$ cat Dockerfile
#Base Image Name
FROM node:20-alpine

#Label names of the project like who is the maintainer project name and project version
LABEL maintainer="abhi@gmail.com"
LABEL project="Demo project"
LABEL version="1.0"

# Arguments that have been set so that can be imported in env variables..
ARG NODE_ENV=production

#Environmental variables that required to run the project
ENV NODE_ENV=$NODE_ENV
ENV PORT=3000

#The project working directory
WORKDIR /app

#copying all the package.json files so that can be used to install all the packages required to run the project
COPY package*.json .

#npm ci-- meaning clean install
RUN npm ci

#Copy the rest of the files of the project to the working directory
COPY . .

#expose the port of the docker container
EXPOSE 3000

#in the terminal run the command node server.js to start the express application
CMD ["node","server.js"]

# #Health check route monitoring every 30s so check whether the app is running and working fine
HEALTHCHECK --interval=30s --timeout=10s \
  CMD curl -f http://localhost:3000/health || exit 1
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program1$
```

Docker build logs

Command: docker build -t program1 .

```
abhi@abhi-ThinkPad-T490: ~/Desktop/DevopsLab/program1

abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program1$ docker build -t program1 .
[+] Building 2.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 1.11kB
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 43.46kB
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY package*.json .
=> CACHED [4/5] RUN npm ci
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:cc810373a9d8d4a273b8753d832022b9d28e8077a71b7cc4ec951a02a876ec0f
=> => naming to docker.io/library/program1
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program1$ docker run --rm -p 3000:3000 program1
Server is Running in http://localhost:3000

```

Running the docker built image in a container and mapping the port

command : `docker run -d -p 3000:3000 program1`

